

Phase 3 Submission Packet

Project Title

Constraint-Aware Fitness Planner

Team Members

Shiwei Jiang

Selected Track

Track B: Interactive artifact

One-Paragraph Project Summary

Constraint-Aware Fitness Planner is a local browser artifact for non-clinical diet and workout planning under real-world constraints such as symptom-trigger histories, dietary restrictions, limited equipment, schedule limits, budget limits, and adherence challenges. The system is agentic because it coordinates structured intake, ambiguity handling, safety-boundary stopping, human prioritization, reviewer-led revision, explicit packet-level handoffs, and state-preserving adaptation through structured post-acceptance execution feedback. The final artifact now delivers a clearer reviewed user plan package with a profile-fit summary, sample meal pattern, grocery focus, session blueprint, expanded weekly schedule, rationale, warnings, next-step guidance, and a downloadable plan-summary export, while keeping the coordination trace and evidence layer visible for grading.

Repository Or Project Link

Current review package for local inspection:

- Clean local repository folder: this repository root

GitHub repository:

- <https://github.com/Shiwei-Jiang/multi-agents-project-constraint-aware-fitness-planner.git>

Video Link

- https://drive.google.com/file/d/1t9Of0rJVdyEZj3Q7Pm51GctNnXfZJSWq/view?usp=drive_link
- Local reference file: `media/demo_video_link.txt`

Final Report

- Source: `docs/final_report.md`
- Exported PDF: `docs/final_report.pdf`
- Coverage: problem and user, architecture, implementation choices, evaluation methodology, results, failure analysis, governance, formal artifact schemas, lessons learned, future improvements, reflection on improvements and next steps, and individual contribution reflection

Architecture Diagram

- Primary Phase 3 architecture and workflow figure: docs/phase3_architecture_workflow.png
- Supporting/legacy architecture figures retained for continuity: docs/architecture_diagram.svg; docs/architecture_diagram_compact.png
- Shows: bounded agent roles, controller-managed branches, handoff packets, stop conditions, and human-in-the-loop checkpoints

Screenshot Index

- Screenshot index file: screenshots/screenshot_index.md
- Coverage summary: landing screen, ambiguity checkpoint, boundary stop, human tradeoff branch, adaptation, uncertainty handling, state/history view, and evaluation/results view
- Screenshot set:
 - screenshots/01_home.png
 - screenshots/02_clarification.png
 - screenshots/03_boundary_stop.png
 - screenshots/04_prioritization.png
 - screenshots/05_adaptation.png
 - screenshots/06_uncertainty.png
 - screenshots/07_state_and_transitions.png
 - screenshots/08_evaluation_outputs.png

Evaluation Summary

- Deterministic core scenarios: 7
- Extended adversarial, regression, and persona cases: 7
- Total evaluated cases: 14
- Full passes: 14
- Partial failures: 0
- Failures: 0
- Full-pass rate: 100%
- Repeated-run stability: 7/7

Interpretation note:

- The 14 / 14 result is intended as evidence of latest-version behavior within the project's bounded Phase 3 scope.
- It is based on internal scenario emulation, saved traces, deterministic reruns, and structured manual review rather than an external user study.
- Historical failures are still preserved in eval/failure_log.md, so the pass table should be read together with the failure-analysis and governance sections rather than as a universal robustness claim.

Primary files:

- eval/test_cases.csv
- eval/evaluation_results.csv
- eval/failure_log.md
- eval/evaluation_scope_note.md
- eval/version_notes.md
- outputs/demo_outputs/evaluation_summary.md

- `outputs/demo_outputs/scenario_trace_index.md`
- `outputs/demo_outputs/end_to_end_trace_baseline.md`
- `outputs/exported_artifacts/automatic_evaluation_snapshot.json`
- `outputs/exported_artifacts/stability_check.json`
- `outputs/exported_artifacts/extended_evaluation_snapshot.json`
- `docs/package_index.md`

Submitted Files And Folders

The submission includes the following files and directories for inspection:

Top-level files:

- `README.md` — project overview, setup instructions, Track B reviewer guide, and evaluation summary
- `AI_USAGE.md` — disclosure of AI tools used, prompts, manual edits, and verification steps
- `platform_notes.md` — environment and platform details

Core system:

- `app/` — browser-based interactive artifact (main Track B deliverable, UI + workflow integration)
- `src/` — core workflow engine (`engine.mjs`) implementing state machine, agents, and coordination logic
- `scripts/` — evaluation runner (`run_evaluation.mjs`) for reproducible testing

Documentation and report materials:

- `docs/` — final report, architecture diagrams, screenshot index, evaluation evidence view, and supporting documentation

Screenshots:

- `screenshots/` — UI and workflow screenshots used as visual evidence in the report

Evaluation package:

- `eval/` — test cases, evaluation results, failure log, scope notes, and version notes

Outputs and evidence artifacts:

- `outputs/` — exported artifacts, scenario traces, evaluation snapshots, and sample runs

Data layer:

- `data/` — manifest describing the no-external-dataset design

Media:

- `media/` — video link file (`demo_video_link.txt`)

Presentation materials:

- `presentation/` — final slide decks and presentation materials

Previous phase materials (for reference only):

- `phase_submissions/` — Phase 1, Phase 2, and Phase 3 submission artifacts

Notes For Final Submission Assembly

- The packet wording now avoids machine-specific filesystem paths and is ready to accept either a final GitHub URL or a compressed project-folder submission.
- The stakeholder wording in the report has been aligned with the Phase 1 feedback so the operational stakeholder is a trainer, coach, licensed dietitian, or referring provider rather than the evaluator.
- The final report now includes an explicit operational risk register and lightweight formal schemas for the main handoff artifacts and session state, which more directly addresses the Phase 2 feedback about governance structure and data clarity.
- The report PDF has been regenerated from the current Phase 3 source.
- The screenshot set has been regenerated after the latest `phase3-v1.6` artifact and evidence update.
- The evaluation exports have been regenerated from `node scripts/run_evaluation.mjs`.
- The final approved project video link is now recorded in `media/demo_video_link.txt`.